

ECE 3441 Spring 2024

Quiz 3

VERSION B Solutions

- Closed book, closed notes
- No cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.
- You have 50 minutes.

Please do not open the booklet unless instructed to do so, or a substantial penalty will result.

First Name Last Name,

Peoplesoft ID,

Date *DD/MM/YYYY*

Question	Points	Maximum
1		35
2		40
3		35

Total: _____/100 possible points + 10 bonus points

Question #1:

A T-V flip-flop works as follows:

When $V = 1$, the next state of the flip-flop toggles.

When $TV = 01$, the next state of the flip-flop is 0.

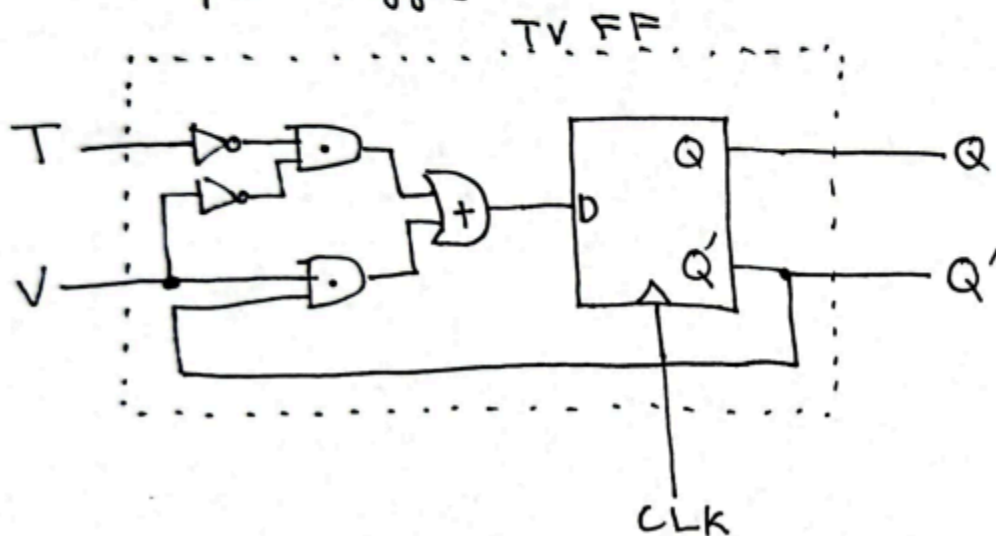
When $TV = 11$, the next state of the flip-flop is 1.

- Find the T-V flip-flop truth table.
- Find the T-V flip-flop characteristic equation.
- Construct the T-V flip-flop from a D flip-flop using the minimum number of NOT gates and 2-input AND/OR gates (Hint: 5 total gates are required)

T	V	Q	Q ⁺	
0	0	0	1	Set
0	0	1	1	Set
0	1	0	1	Toggle
0	1	1	0	Toggle
1	0	0	0	Reset
1	0	1	0	Reset
1	1	0	1	Toggle
1	1	1	0	Toggle

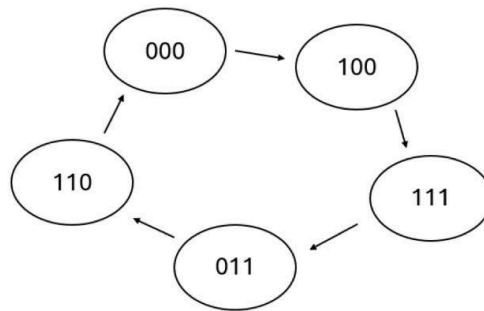
Q ⁺	T	V
0	0	1
1	0	0
0	1	0
1	1	1

$$Q^+ = T'V' + VQ'$$



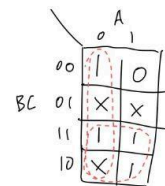
Question #2:

Design a counter based on the following sequence using T flip-flops. Your design should include a circuit drawing and every k-map should be simplified to a minimum SOP. There may be multiple solutions. What is the next state for each of the unused states? Do the unused states lead back into the sequence?

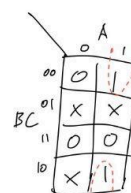


ABC	$A^+B^+C^+$	$T_A T_B T_C$
000	100	100
001	xxx	xxx
010	xxx	xxx
011	110	101
100	111	011
101	xxx	xxx
110	000	110
111	011	100

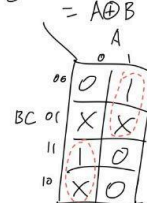
$$T_A = A' + B$$



$$T_B = AC'$$



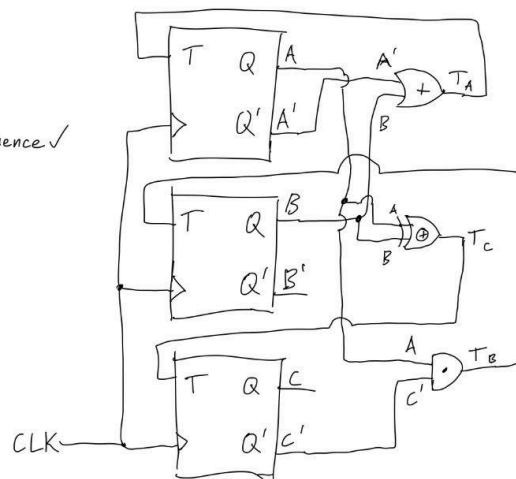
$$T_C = AB' + A'B = A \oplus B$$



UNUSED:

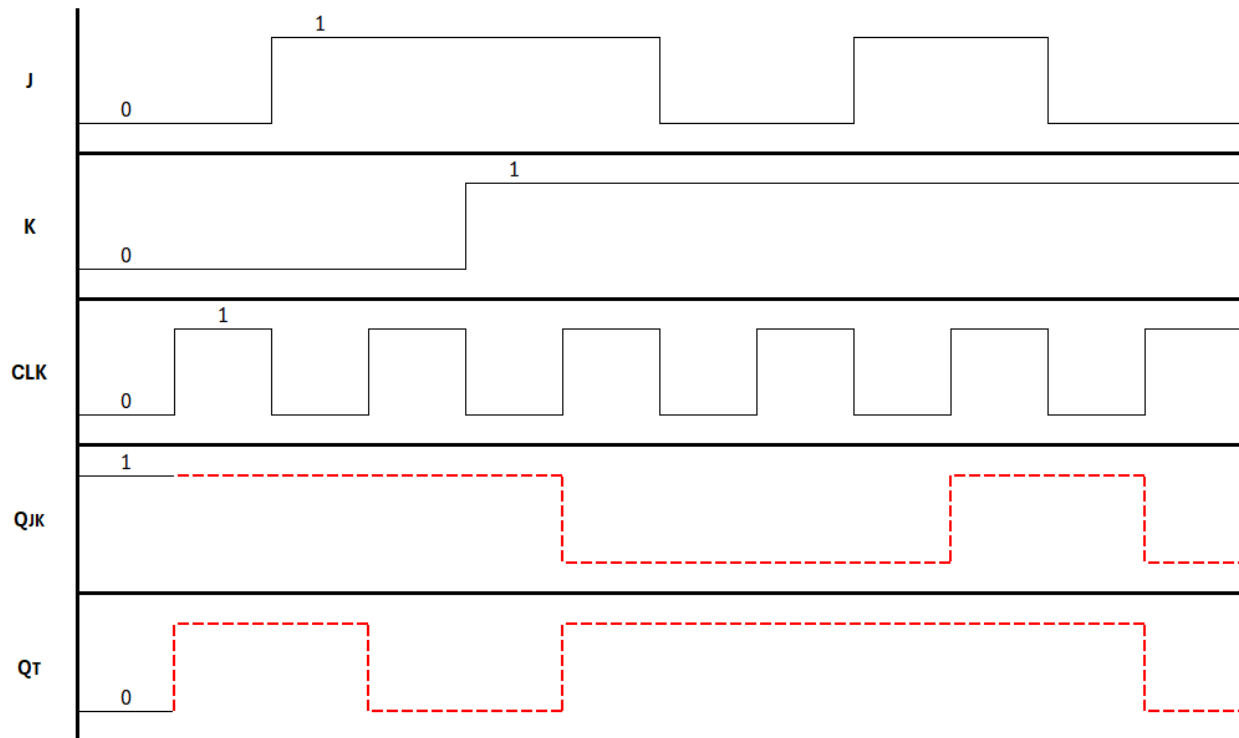
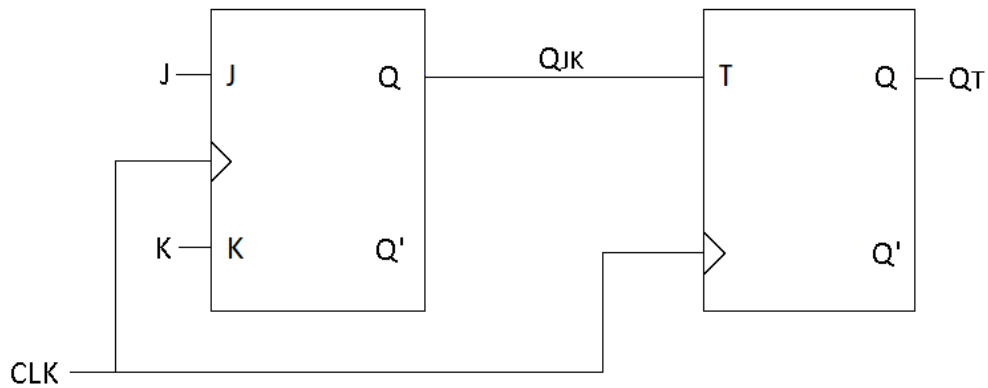
ABC	$T_A T_B T_C$	$A^+B^+C^+$
001	100	101 to 101, then on sequence ✓
010	101	111 on sequence ✓
101	001	100 on sequence ✓

This counter is secure.



Question #3:

Given this circuit containing one JK flip-flop and one T flip-flop, complete the timing diagram. The starting states for Q_{JK} and Q_T are 1 and 0, respectively.



T FLIP-FLOP



T	Q	Q^+
0	0	0
0	1	1
1	0	1
1	1	0

$$Q^+ = T'Q + TQ' = T \oplus Q$$

JK FLIP-FLOP



J	K	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$$Q^+ = JQ' + K'Q$$

SR FLIP-FLOP



S	R	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	-
1	1	1	-

$$Q^+ = S + R'Q \quad (SR = 0)$$

D FLIP-FLOP



D	Q	Q^+
0	0	0
0	1	0
1	0	1
1	1	1

$$Q^+ = D$$

S-R Latch

